PulmoTwin: 48-Hour Respiratory Digital Twin

System Architecture & Multi-Modal Pipeline • Monalika Doda, SRM University Delhi-NCR

1. INGESTION & COHORT SYNTHESIS LAYER

Synthetic Patient Phenotypes
- 500 synthetic COPD/asthma patients
- Uses GOLD/GINA stage categories
- Proportions are our assumptions
- Zero real patient data (no PHI)

365-Day Synthetic Wearable Telemetry
- 182,500 patient-days simulated
- SpO2, HR, RR, Sleep, Steps
- Individual baselines + AR(1) noise
- 2-3 day pre-flare biomarker drift

Real Modelled Weather & Air (Open-Meteo)
- CAMS PM2.5, PM10, Dust -> CPCB NAQI
- ERA5 Reanalysis: Temp, Swing, Wind
- CC BY 4.0 Data Attribution
- Illustrative seasonal pollen curve

2. TIME-SERIES FEATURE ENGINEERING & ENVIRONMENTAL FUSION

Biometric Rolling Features (17 feat)
- Deviation from personal baseline
- 3-day rolling trend (SpO2, HR, RR)
- Exertional drop ratio (Steps/base)
- Sleep duration deficit

Environmental Exposure (29 feat)
- Delhi CPCB NAQI mean & 3d max
- Pre-monsoon dust peaks
- Diurnal temperature swings (ΔT)
- Rain-attenuated seasonal pollen

Patient-Level Train/Test Partition
- GroupShuffleSplit: 400 train / 100 test
- Zero cross-patient leakage
- Missing sensor dropout test
- Compact dataset: 62.98 MB tabular

3. PREDICTIVE AI ENGINE & SHAP ATTRIBUTION

XGBoost Binary Classifier
- 48h Advance Flare Forecast
- ROC-AUC: 0.9982 | PR-AUC: 0.9078
- Day Recall: 93.4% | Prec: 75.6%
- 4-Stage Ablation Benchmarks

SHAP Factor Attribution
- TreeExplainer sequential attribution
- Ordered by factor size (|phi|)
- Model Baseline (36.2%) + Bars = Risk%
- 9 Clinical actionable factors

What-If Sensitivity Engine
- Recalculates model output
- Air filtration (AQI delta)
- Inhaler compliance booster
- Model sensitivity on synthetic data

4. CLINICAL DECISION SUPPORT & DASHBOARD INTERFACE

FastAPI REST Backend
- Asynchronous API (Port 8000)
- 365-day Timeline Replay
- Patient list sorted by 48h risk
- OpenAPI / Swagger docs

React Clinical Dashboard
- 3D breathing lung twin (Three.js)
- Light/glass telemetry card
- 5 Vital tiles with 7d sparklines
- Error boundary protected

Workload & Episode Grouping
- Consecutive alert days grouped
- Delhi map with patient pin
- 1-Click triage action orders
- 0.77 alert days/patient/month